Auditing and Partially Reproducing the SLRTP2025 Back-Translation Evaluator

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Abstract

We audit whether the released SLRTP2025 back-translation (BT) evaluator can be reproduced from its public artifacts, an evaluation-only repository without training scripts. A line-level audit of our own reconstruction code identified seven defects relative to the released configuration's semantics; the corrected re-implementation recovers most of the evaluator's in-domain competence (eight seeds: dev BLEU-4 11.59–12.62, mean test-split decoding 12.27 $\pm$ 0.46, versus the released 13.38 / 12.78).

Two qualitative signatures nevertheless fail across all 69 non-degenerate runs—runs that share one code base, one data pipeline, and one undisclosed author inference (gradient-clip threshold 1.0): the training-pool memorisation signature (released 78.8 BLEU / 70.7% exact match; constructible ceiling 12.08 / 2.5%) and a positive response to a constructed adversarial replay probe (released **+10.24**, post-selected). A clipping sensitivity grid resolves both in our reconstruction: with clipping disabled—the candidate upstream framework's default when, as in the released configuration, no clip field is specified—the reconstruction reproduces both signatures (readout 99.98 / 99.9%; probe gap **+11.19**), consistent across the three tested seeds and on a secondary dev-split consistency check (**+8.0** to **+8.6**; released **+8.10**). Readout and gap vary monotonically across the clip ladder; the released operating point's individual coordinates are bracketed by different thresholds but not matched jointly, so we claim qualitative signature reproduction and coordinate-wise bracketing—not numerical reconstruction. Update-level diagnostics show the clipping effect operates through the training dynamics (Adam moments, weight decay, loss-scale interaction), not through proportional shrinkage of the parameter update, and we do not claim confirmation of the released run's original training mechanism. The probe response is a case-specific adversarial unit test, not a general BT finding; the transferable contributions are the artifact-audit protocol and the lesson that an undocumented framework default can make reproducible properties look unreproducible.

Keywords: sign language production, back-translation evaluation, reproducibility audit, reference-frame sensitivity, learned evaluation metrics, PHOENIX-2014T

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit asks to what extent the released SLRTP2025 BT evaluator can be reproduced from its public artifacts by our own best-effort implementation, and what stood in the way where it could not. We address two research questions:

RQ1. To what extent can a best-effort, artifact-conditioned implementation of the released training configuration—with every disclosed defect corrected and every remaining inference bracketed by sensitivity runs—reproduce the released evaluator's measurable properties (in-domain competence, training-pool readout, and probe response)?

RQ2. How sensitive is the constructed probe's response to the reference text and to donor origin?

Beyond the two research questions, we report a *descriptive observation* (not a research question): the two properties that initially failed to reproduce—the training-pool memorisation signature and the probe response—are blocked not by an unrecovered training mechanism but by a single author inference in our own reconstruction. Across the 69 non-degenerate constructible runs, all of which share a gradient-clip threshold of 1.0, the readout stays at or below 12.08 BLEU (2.5% exact match) versus the released evaluator's 78.8 / 70.7% and every probe gap is negative; yet the most competent of them (the seven-correction family) already reach the released evaluator's test-split decoding competence (family mean 12.27 vs. 12.78), so the shortfall is not surface competence. The sensitivity grid of §4.3 then shows both properties to be a *clipping-regime* effect in our reconstruction: with clipping disabled—the candidate upstream framework's default when the released configuration specifies no clip fields, as it does—the reconstruction reproduces both signatures, with intermediate thresholds interpolating monotonically between the regimes. Throughout we distinguish qualitative signature reproduction (demonstrated) from numerical reconstruction of the released operating point (not achieved) and mechanism confirmation that the released training itself ran unclipped (not established; §4.3). The replay probe itself is an *adversarial lexical-coupling stress test*: the source-conditioned retriever selects donors by surface vocabulary overlap with the German source, which is identical to the scoring reference in PHX-public, so the probe deliberately targets the strongest form of the shortcut rather than a deployable retrieval pipeline.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026) (the latter a preprint); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai et al. (2025); Jiang et al. (2025), yet BT remains the standardized infrastructure on which the SLRTP2025 ranking depends Walsh et al. (2025). Chen et al. (2022) showed that learned metrics can fail to reproduce due to undocumented preprocessing and missing code—an anatomy directly relevant here, where seven defects in our own reconstruction were identified and corrected. What prior audits of PHOENIX-2014T have not shown is whether the released evaluator itself is reproducible from its artifacts: under it, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU versus 12.78 for the recorded test poses, and the question is whether that response and the evaluator's training-pool behaviour can be rebuilt from the public release.

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so the adversarial probe isolates a risk that production systems face in diluted form; the transferable parts of the audit are the protocol (provenance table, checkpoint registry with SHA-256 deduplication, evaluator-replication controls, teacher-forced-vs-free-decode diagnostics), while the specific probe response is a PHX-public case study that does not support general claims about BT evaluation or retrieval-based SLP.

2 Related Work

Note on citations and preprints. Two 2026 PHOENIX-2014T audits we build on are peer-reviewed: Czehmann et al. (2026) (LREC 2026 12th Workshop on the Representation and Processing of Sign Languages) and Alkain et al. (2026) (Frontiers in Artificial Intelligence), as is Artiaga et al. (2025) (Findings of EMNLP 2025). Where we cite arXiv preprints Hong and Košecká, Jana (2026); Cory et al. (2026); Desai et al. (2024) (marked "Preprint" in the bibliography), we treat their findings as preliminary reports rather than peer-reviewed results; conclusions in this audit do not hinge on the correctness of any single preprint. All arXiv citations were accessed on 2026-08-10.

Learned evaluation and its reproducibility.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); Saunders et al. (2021) proposed BT as an SLP evaluation protocol, arguing it shifts absolute scores but not rankings between similar configurations—a condition unmet in our family. SignBLEU Kim et al. (2024), SiLVERScore Imai et al. (2025), and human-correlated metrics Jiang et al. (2025); Cory et al. (2026) offer complementary views, but BT BLEU remains the standardized infrastructure on which the SLRTP2025 ranking depends Walsh et al. (2025); auditing its reproducibility is infrastructure maintenance, not a metric-superiority claim. Chen et al. (2022) showed that reproducing BERT-based evaluation metrics can fail through undocumented preprocessing, missing code, and baseline differences even when weights are released; our audit targets a full evaluator pipeline and adds a resolved case—the failure traced to an undisclosed author inference, not to the artifacts (69 non-degenerate decoded runs, all with strictly negative gaps, all sharing clip 1.0; 68 distinct SHA-256 weight binaries after one byte-identical collision).

Retrieval, shortcuts, and corpus audits.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025's leading entry is retrieval-based Walsh et al. (2025); memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020), and model-based generation metrics are criticized by Yazdani et al. (2026); Cory et al. (2026); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. (2026) released human back-translations and documented reference information loss, and Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora; Artiaga et al. (2025) likewise showed that signer-dependent evaluation overestimates SLT capability. None of the three runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

PHOENIX-2014T as a language resource: known limitations.

PHOENIX-2014T Camgoz et al. (2018) (8,257 weather-forecast interpretations by 9 signers) is the most widely used DGS parallel corpus, but four properties constrain generalizability: (i) template density inflates train–test n-gram overlap Alkain et al. (2026); (ii) reference texts are speech transcriptions, not DGS translations Czehmann et al. (2026); (iii) the 178-keypoint pose materialization encodes joint centres and coarse facial geometry but not appearance texture, hand–body contact, occlusion-robust depth, or reliable eye gaze (our MSKA control uses 133 keypoints; OR1 Sup. L); (iv) signer-dependent evaluation can overestimate capability Artiaga et al. (2025). The lexical coupling between source and reference amplifies these effects in our adversarial probe; the transferable audit components (provenance table, checkpoint registry, leakage controls) apply to any learned evaluator paired with a public training corpus.

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint, $o_{i,n}$ the pose sequence for output condition i and item n, and y_n the reference text; E_c produces hypotheses $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where M is the corpus-level sacreBLEU functional unless stated otherwise. We call a result a *descriptive non-reproduction* when a probe response observed under the released evaluator does not appear across independently constructed checkpoints under a fixed output set; the term is descriptive and does not imply a causal checkpoint-identity effect. The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

Competence comparison.

Competence is reported throughout as *continuous* metric comparisons against the released evaluator (e.g. seven-correction dev BLEU-4 11.59–12.62 and train readout 11.45–12.08 vs. released 13.38 and 78.8). An author-operational pass/fail gate and its threshold-insensitivity grid are relegated to the supplement (OR1 Sup. D–E) and are not used as primary evidence in the main text.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences, against the official PHOENIX-2014T split sizes of 7,096/519/642: 36/4/1 official IDs are absent from the released pose materialization, and every result is conditioned on the complete release. An ID-level reconciliation (`results/split_reconciliation.json`; OR1 Sup. A) classifies all 41 missing IDs—each is present in an independent upstream skeleton extraction, so the drop occurred inside the SLRTP2025 pose packaging; the dropped items show no duration, signer, or template-density anomaly, and the evaluator bundle's shipped `train.pt` contains exactly 7,060 items, so the memorisation signature we audit is defined over the same 7,060 items the released evaluator trained on. Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; OR1 Sup. L); the protocol summary is in OR1 Table S38.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution. The data flow is: official PHOENIX-2014T video at 25 fps $\rightarrow$ SLRTP2025 skeleton extraction at 25 fps $\rightarrow$ our `[::2]` subsampling to 12.5 fps for evaluator input; the official evaluation command uses `--fps 25` (the input file's native rate), but the evaluator was trained on 12.5 fps-subsampled poses, so our subsampling matches the training distribution rather than the command-line flag. A three-path regression confirms no double subsampling (details in OR1 Sup. F). PHX-public uses α=0.88 (selected on dev); the supplementary splits use α=0.5.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is `9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbefa301`. Reproduction is supported at four layers (OR1 Sup. Q, Table S44): **L1** recompute selected reported statistics (`make core-audit` regenerates the registry and verifies headline numbers from committed per-item decodes; it does not retrain models or decode from checkpoints); **L2** analysis re-execution from committed per-checkpoint data; **L3** re-decoding fixed outputs from checkpoint weights (requires the training bundle); **L4** from-scratch retraining (restricted PHOENIX-2014T licence, exact environment, ≈200 GPU-hours). A claim-to-file manifest mapping each headline number to its source file and verification command is in `claim_manifest.json` (artifact root); the per-claim reproducibility accounting is in OR1 Sup. Q and Table S50, and the L1/L2 commands exit non-zero on any mismatch (run transcripts in the artifact). A pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in OR1 Sup. N.

Lexical coupling of the source-conditioned probe (adversarial design).
In PHX-public the source sentence is identical to the reference text at the sequence level, and the TN-PURE-v1 retriever selects donors by token-set Jaccard similarity to that same text, so the probe deliberately targets the strongest form of the lexical-coupling shortcut: the +10.24 gap is the expected output of an adversarial design, not a discovery about ranking behaviour, and the human-reference analysis of §4.3 therefore mixes reference-text properties, coupling disruption, paraphrase shift, and BLEU floor effects (full framing in OR1 Sup. T).

Retrieval composition and donor-origin probe systems.
TN-PURE-v1 uses whole-donor copy at 12.5 fps; TN-PTCOMP-v1 composes a PT scaffold with donor hands and face via α-blending (α=0.88; joint-set definitions in the artifact). Frozen donor-origin probes are **UNSEEN-PURE-v1** (retrieval from the 640 other test poses), **CROSSFIT-PURE-A/B** (hash-split folds), and **SEEN-PURE-MATCHED-v1** (a train-pool donor within ±0.1 Jaccard of each UNSEEN donor; 622/641 admit one); all are constructed probes, not deployable generators (construction rules in the artifact; OR1 Sup. E).

3.3 Evaluator checkpoint family

The main inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and 36 *recipe-conditioned* runs trained from scratch with the documented architecture on a common frozen partition: the eight *seven-correction* runs (§3.4; seeds 42–49), fourteen *partially-corrected* runs (4 validation-freq-misread + 10 step-corrected), and fourteen *legacy* runs (6 prospectively specified seeds 101–606 + 8 extension seeds 707–1405). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps, with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.3, and intermediate checkpoints are not published. All recipe-conditioned runs share the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, wd .001), batch size 256, and plateau ×0.8 LR schedule (full provenance in OR1). In total 76 from-scratch training runs across 12 families (registry in OR1 Sup. D); 74 were beam-3 decoded under the canonical donor registry (73 unique weight binaries after one byte-identical collision), five degenerate runs excluded—leaving 69 non-degenerate decoded runs, all with strictly negative gaps. All 69 share code, data, architecture, and the clip-1.0 inference (§4.3); they are coverage evidence, not independent replications.

What the 76 runs are—and are not.

The 76 runs are *not* 76 same-recipe replication attempts: the family-provenance table (OR1 Sup. P) maps each family to its public source, known deviation, checkpoint-selection rule, and dev BLEU-4 range. The eight seven-correction runs implement the released config to the maximum extent the artifacts identify, with two disclosed author inferences bracketed by sensitivity runs; the partially-corrected and legacy families corroborate, and the remaining 40 runs are diagnostic, sensitivity, or competence-extrapolation analyses. The non-reproduction conclusion rests primarily on the seven-correction family.

Training-pool readout (descriptive observation).

We define *training-pool readout* as the in-sample autoregressive free-decode BLEU and exact-match (EM) rate on the full 7,060-item training set under beam-3 decoding—not teacher-forced accuracy. The released evaluator reads out its training pool at 78.8 BLEU / 70.7% EM while retaining dev/test 13.38/12.78; the canonical constructible families read out at 6.6–12.08 BLEU (1.3–2.5% EM), whereas the unclipped runs exceed the released values (99.98 / 99.9%). The readout-to-dev ratio (released ≈5.9; canonical family 0.93–1.12; unclipped ≈6.1) summarises how far a model decodes above its in-domain performance. These are continuous descriptive observations; any threshold is post-hoc.

3.4 Reconstruction sanity checks and training diagnostics

Three checks on the recipe-conditioned families rule out basic plumbing failures (data loading, masking, decoding alignment, vocabulary mismatch, pose-subsample direction): a small-sample overfit test (50 sequences reach free-decode BLEU 100.00), a unified-inference regression (the released checkpoint reproduces canonical REC 12.78 exactly through one pipeline with byte-identical vocabularies), and the teacher-forced/free-decode split (released: 96.5% train token accuracy and 78.8 train free-decode BLEU vs. defective families' 55–60% and 7.2–9.3). These checks do *not* rule out subtler training-objective, normalisation, or scheduler defects—indeed seven such defects were later found; training logs confirm normal optimisation for the defective families, and a 25→12.5 fps subsampling regression test passes (OR1 Sup. R).

Seven implementation defects, and the seven-correction re-implementation.

A line-level audit of our own training scripts identified *seven* defects relative to the released `config.yaml`'s semantics—four found by forensic arithmetic against the released training log, three more by line-level audit of the surviving code:

1. *CTC loss omitted* (v1) despite `recognition_loss_weight=1.0`.
2. *Validation frequency misread* (v1): `validation_freq=14` is 14 optimiser *steps* (JoeyNMT semantics; the released log validates at steps 14, 28, ...), not 14 epochs.
3. *Selection branch never saved* (v1): the `--selection bleu` branch had `improved` hard-coded `False`.

4. *Non-autoregressive selection decode* (v1.5/v2): per-position argmax over teacher-forced logits instead of autoregressive decoding; v3 corrected this to beam-3.
5. *Validation mask inversion* (v1–v3): `txt_mask=True` marks *valid* positions, but token counts used the negated mask, counting padding.
6. *Loss-normalisation mismatch* (v1–v3): the config specifies `translation_normalization: batch`; our code used per-token mean CE. The released log pins the logged semantics, and the training-time reduction (token-sum CE divided by sentences) is an inference corroborated by train-loss magnitudes and bracketed by sensitivity runs (§4.3); under Adam this changes the effective loss scale by the mean target length ($\sim 15\times$), altering both the gradient-clipping regime and the relative strength of weight decay.
7. *Stopping-rule mismatch* (v1–v3): the released log refutes patience-based early stopping outright—the released run improved last at validation #130 (step 1820) and then ran 72 further non-improving validations to step 2828, an external termination; the only stopping condition reproducible from the config is the learning-rate floor (1e-8).

The seven-correction re-implementation (`train_faithful.py`) corrects the seven identified defects and takes every remaining parameter from the released config: seed 42 (the released seed) plus seeds 43–49 for within-implementation dispersion; joint CTC+translation with batch normalisation; step-level validation every 14 optimiser steps; autoregressive beam-3 selection identical to the evaluation decoder; a plateau scheduler whose 20 decays match the released log's cadence field-for-field; the LR-floor stopping rule; and gradient clipping at 1.0 (disclosed as an author inference—the config does not specify clipping; with batch-normalised loss the clip genuinely binds, pre-clip gradient norms over the first ten steps being 377, 568, 145, 33, ...). A 56-step smoke run reproduces the released log's step-14 dev token-sum NLL to within 1.1% (`results/faithful_smoke_gradnorms.json`). The one released-log column we cannot reproduce is the `Recognition Loss` magnitude: our step-14 dev CTC is 39.9 per sentence, 5.5 per gloss token, 0.74 per input frame, and 20,527 as a raw sum, and none matches the logged 17.28, so the normalisation of the original CTC term is not identifiable from public artifacts. The two remaining author inferences (CTC normalisation and clip threshold) and the stopping-rule difference are therefore bracketed by sensitivity runs in §4.3. The bracketing resolves them asymmetrically: the CTC normalisation proves non-identifiable but mild, whereas the clip threshold proves *load-bearing*—the released config's omission of clip fields, read against the candidate-upstream default of no clipping, makes our 1.0 an eighth defect (§4.3, Table 2). In practice each seven-correction run terminated at the plateau scheduler's *numerical* floor (LR frozen at 4.4e-8), with the best checkpoint unchanged for the final 218–338 validations; the released run itself ended mid-schedule at step 2,828, so §4.3 additionally evaluates checkpoints fixed at the released run's best (1,820) and final (2,828) steps. The implementation ladder from the defective protocols to the unclipped resolution is OR1 Table S53, and the config-mapping table (OR1 Sup. F) maps every load-bearing field to the original-framework semantics, our v1–v3 semantics, and the seven-correction semantics, with the evidence source for each field.

3.5 Metrics, uncertainty, and discovery timeline

We report corpus BLEU-4 on the 0–100 sacreBLEU scale Post (2018) (sacrebleu 2.5.1; signature in OR1 Sup. Q), with BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020), and WER as cross-metric checks (all share decoded outputs) and teacher-forced BT likelihood as a non-decoded control. Uncertainty uses donor-cluster bootstrap (primary; 10,000 resamples; 565 unique donors; OR1 Sup. G) and item-level bootstrap; clustering sensitivity for the headline gap is tabulated in OR1 Sup. G (the signer and broadcast-show designs are exploratory references only). Seed dispersion is reported as descriptive Student-t intervals (OR1 Sup. B); the six primary seeds give $[-2.48, -0.28]$. The headline gap is a **post-selected descriptive estimate**: the +10.24 response was first observed on 2026-07-24, the Jaccard retriever, PURE protocol, and 641-query set were locked before reconstruction training began, and six legacy seeds (101–606) were prospectively specified before inspection, but the headline cell was selected as the max-gap cell among the explored probe cells (§4.1); all other contrasts are Holm-corrected within a defined family or labelled descriptive (full provenance in OR1 Sup. C).

3.6 Terminology and claim hierarchy

Throughout, *run* denotes a single training execution; *checkpoint* the selected weights from that run; *unique weight binary* counts distinct SHA-256 hashes (one validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact); *family* a group of runs sharing a training protocol; *evaluator* a checkpoint used to decode fixed outputs. Claims use four evidence tiers: Descriptive (D; no formal error control), Sensitivity (S; explicit post-hoc condition), prospectively specified within-project follow-up (C; only the six legacy seeds 101–606, seed-level variation rather than independent replications), and Not Identified (NI). Checkpoint families stratify into *Seven-correction* (8 runs, seeds 42–49; primary evidence for RQ1), *Partially-corrected* (14), *Secondary* (14 legacy seeds, 6 prospectively specified), and *Diagnostic* (40); the full registry, family table, the auto-generated accounting table (76 trained / 74 decoded / 73 unique / 69 non-degenerate, all 69 gaps negative), and the claims register (estimand, unit of analysis, selection status, uncertainty, and evidence tier per headline claim, so that descriptive, confirmatory, item-level, and seed-level quantities are not read interchangeably) are in OR1 Sup. D and OR1 Tables S26–S27.

4 Results

4.1 Adversarial probe response: an attack result, confirmed and localised

Under the released SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses—a difference of +10.24 BLEU points. We present this as the output of the adversarial unit test defined in §3: the number measures the evaluator's susceptibility to this specific lexical-coupling shortcut, not general retrieval quality or signing adequacy. Because the headline cell was selected as the maximal-gap cell among the explored probe cells (OR1 Table S46), we do not attach a confirmatory-looking interval to it; its resampling intervals are reported in OR1 Sup. G as post-selection descriptive quantities.

Three results characterise the attack precisely. **(i) Secondary dev-split consistency check:** the identical probe construction applied to the development split (515 items, never used for probe construction or exploration) reproduces the response at consistent magnitude—REC 13.38 (matching the released training log's 13.38 to three digits, independently validating the decode pipeline) vs. PURE 21.47, a gap of +8.10 against the test-split +10.24 (OR1 Sup. S); the unclipped reconstructions reproduce the same dev-split response at matching magnitude (§4.3). This is a consistency check, not an independent held-out confirmation: the dev split is used for checkpoint selection and LR scheduling in every reconstructed run, and 13.38 is itself the released run's selection metric. **(ii) Text-free selection rules do not reproduce it:** a pose-similarity retriever that never touches text scores 0.45 (gap −12.33), indistinguishable from random donors (−11.88) and far below text-retrieval TN-PURE (23.02). Because such rules also retrieve less lexically similar donors, they conflate loss of coupling with loss of retrieval quality; the retrieval-query × scoring-frame grid on the 461-item human-reference subset separates the two (OR1 Sup. T): retrieving with the human back-translations while scoring against the original references removes the response (-4.02 $[-5.50, -2.62]$) although the donors still average Jaccard 0.30 to the originals, whereas re-coupling query and scoring frame on the human texts partially restores it ($+4.29$ $[3.09, 5.49]$; original diagonal +10.03, crossed cell +0.95). Within the tested grid the response coincides with lexical identity of retrieval query and scoring reference, not with text-based selection as such; randomisation within the Jaccard top-k interpolates between regimes (+7.06 at k=5, +3.34 at k=20). **(iii) Deployment-realistic systems do not show it:** a noisy-gloss retrieval pipeline scores only −0.13 relative to REC under the same evaluator (§4.5).

Further controls localise the effect to source-conditioned donor retrieval (retriever comparison, oracle-gloss and PT-composition baselines, teacher-forced NLL, complementary text metrics, decoding-parameter sweeps; OR1 Table S32, OR1 Sup. F): pure donor copy (23.02) exceeded PT-composed (18.86), oracle-gloss (11.4–11.8), PT baseline (1.59), and 20-seed random donors (0.90/0.77); source-text Jaccard was the strongest retriever (LaBSE 19.2, multi-SBERT 16.5); beam/length sweeps leave the gap at +9.9 to +10.8.

Probe multiplicity and post-selection status.

Before freezing the headline probe we explored four retrieval strategies crossed with three construction variants on the released evaluator alone (6 of 12 cells explored, plus two random baselines; OR1 Table S46) and selected the maximal-gap cell; no multiplicity correction would make a post-hoc selection confirmatory, so the headline gap carries no confirmatory interval. The six prospectively specified legacy seeds (§3.5) provide the only prospectively specified evidence (for the non-reproduction direction, not the gap magnitude).

The headline unit-test figure (released response vs. the all-negative canonical panel, with donor-cluster bootstrap CIs) is OR1 Fig. S7.

The probe reversal motivates the prior question: can the public recipe reproduce the released evaluator at all?

4.2 RQ1: What the seven-correction re-implementation recovers, and what one inference still hid

Table 1 reports the seven-correction family (eight seeds, §3.4) against the released evaluator on four dimensions: dev competence, test-split decoding of the recorded poses (the probe panel's own REC baseline), full-training-pool readout, and the probe gap. Three findings follow; the second and third describe the position *before* the clipping resolution of §4.3, which completes the reconstruction.

Table 1: The seven-correction family (`train_faithful.py`, seeds 42–49; the seven identified defects corrected) versus the released evaluator. Dev/test BLEU-4 are uniform beam-3 decodes (515/641 items); readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate; gap is the canonical PURE−REC sacreBLEU-4 difference on the 641-item test panel. The released evaluator's row uses the same protocols.

Seed	Dev BLEU-4	Test REC	Train readout (EM%)	PURE	Gap
42 (released seed)	12.05	**13.11**	11.96 (2.4%)	10.32	−2.80
43	11.59	11.60	11.45 (2.3%)	10.22	−1.37
44	11.95	12.39	11.74 (2.3%)	10.13	−2.25
45	**12.62**	12.04	11.87 (2.5%)	10.08	−1.97
46	11.80	12.26	11.48 (2.3%)	10.16	−2.10
47	12.03	12.53	12.08 (2.4%)	10.34	−2.20
48	12.17	12.33	11.67 (2.4%)	9.98	−2.35
49	11.93	11.86	11.48 (2.4%)	9.77	−2.09
Released evaluator	13.38	12.78	78.8 (70.7%)	23.02	+10.24

Finding 1 (surface competence largely recovered).

The seven-correction runs reach dev BLEU-4 11.59–12.62 (primary seed-42 run: 12.05) against the released 13.38—closing most of the gap that the seven defective implementations left open (their ceiling: 10.5). On the test split, where the probe panel itself is scored, the seven-correction evaluators decode the recorded poses at mean 12.27 ± 0.46 SD (range 11.60–13.11, n=8) against the released 12.78: the family mean sits 0.51 BLEU below the released evaluator, the best seed (42, the released seed) exceeds it by 0.33, and we report the mean and range—not the best seed—as the reconstruction evidence, since a best-of-8 statistic is optimistically biased. The earlier failure to approach the released evaluator's competence is therefore explained *predominantly* by the seven implementation defects; the residual gap and the signature below are explained by the eighth, the clip-threshold inference (§4.3), which raises dev to 16.32–17.17 when corrected or relaxed.

Finding 2 (training-pool readout signature blocked by one inference).

What the seven-correction family does not reproduce is the released evaluator's training-pool behaviour: full-pool free-decode readout is 11.45–12.08 BLEU with 2.3–2.5% exact match, against the released 78.8 BLEU / 70.7% EM. The residual gap is therefore concentrated in a *memorisation signature*—near-perfect decoding of the evaluator's own training pool at retained in-domain generalisation—and not in surface competence. Five leakage controls (OR1 Sup. M) rule out teacher-forcing, output–reference misalignment, pose-ignorant lookup, and narrow-text memorisation shortcuts for the released evaluator's readout, so the signature reflects pose-conditioned training-pool exposure. This shortfall is not an unrecovered training mechanism: §4.3 shows it to be a clipping-regime effect that the unclipped default of the underlying framework reproduces (readout 99.98 BLEU / 99.9% EM across seeds), with the released evaluator's 78.8 / 70.7% falling between clip 3.0 (53.21/31.8%) and clip 5.0 (82.69/68.4%).

Finding 3 (probe response blocked by the same inference).

All eight faithful evaluators give negative probe gaps (-2.80 to -1.37; Table 1), including the seed-42 and seed-47 runs whose test REC matches or approaches the released 12.78 (13.11 and 12.53). The non-reproduction of the adversarial probe response therefore cannot be attributed to uniformly lower surface competence. Within the clip=1.0 constructible family the readout–gap association is in fact *negative* (Spearman ρ=-0.56 over the 37 constructible checkpoints with uniform readouts and canonical gaps; ρ=-0.44 with the released evaluator included), and the released evaluator is the single positive-gap checkpoint of that panel, lying far outside the family's readout range (6.6–12.08 vs. 78.8). The clipping sensitivity (§4.3) resolves the apparent paradox: the readout–gap relation reverses across clipping regimes—strictly negative within the clip=1.0 family, strictly positive along the clip ladder—so no amount of seed variation at clip 1.0 could have produced the response.

Descriptive triangulation for the constructible family (Fig. 1b,c; OR1 Sup. D): across the 38 constructible checkpoints with uniform readouts (37 of which have canonical gaps), readout is associated with competence within the family (ρ=0.86–0.88) and *negatively* with the gap (ρ=-0.56; -0.44 with the released included)—but all 38 share clip 1.0, and §4.3 reverses the association across regimes (unclipped: readout 99.98, gap +11.19), so the within-regime correlation must not be read as "memorisation antagonises the probe". No distillation student transfers the readout pattern (OR1 Table S36); the reversal is robust to weight noise near the released checkpoint ($\sigma \leq 3 \times 10^{-3}$; nine fine-tunes), which tests the released weights' vicinity, not an independent construction. Three-view summary: OR1 Table S49.

Leakage checks, distillation, and extrapolation.

Five controls (OR1 Sup. M) rule out four shortcut classes for the 78.8/70.7% readout—teacher forcing, output–reference misalignment (label permutation collapses BLEU to ∼1.0), pose-ignorant lookup (input-side pose permutation collapses BLEU to <1; a same-signer held-out control reaches only 12.19), and narrow-text memorisation—consistent with pose-conditioned training-pool exposure. Fifteen distillation students all have negative gaps and transfer no readout pattern (OR1 Table S36); competence extrapolation and per-item decompositions are in OR1 Sup. K. In summary, within the clip=1.0 panel the released evaluator is the only checkpoint combining high training-pool readout with retained in-domain performance; the seven-correction family closes the surface-competence dimension, and §4.3 closes the remaining dimension.

4.3 Sensitivity: stopping rule, CTC normalisation, and gradient clipping

The seven-correction protocol differs from the released run in three identified-but-unverifiable respects: stopping point (LR-floor termination vs. step 2,828), CTC-term normalisation (unidentifiable from the released log; §3.4), and the clip threshold (author inference). Three sensitivity experiments bracket these degrees of freedom; all runs live outside the canonical 76-run panel (separate registry; `results/faithful_steps_eval.json`). The first two leave §4.2 unchanged; the third overturns it and resolves the non-reproduction: *the gradient-clip threshold is load-bearing*, and with clipping disabled—the candidate-upstream default when the released config omits clip fields (§4.3)—the reconstruction reproduces both signatures.

Step-matched checkpoints (stopping rule).

Fresh seven-correction runs (seeds 42–44, identical protocol) save checkpoints at the released run's best (1,820) and final (2,828) optimizer steps, so the comparison is fixed-step rather than fixed-protocol-termination (OR1 Table S51). The fresh runs replicate the published family—seeds 42 and 43 *byte-identically* (their selected `best.ckpt` files carry the published runs' SHA-256 hashes) and seed 44 in best dev BLEU-4 (12.05 / 11.59 / 11.95 across seeds 42–44, matching Table 1)—so the step-matched checkpoints are the published trajectories' own states at those steps, and the same-seed retraining doubles as an L4 determinism check. At the released run's own best step, the reconstructions read out 10.85–11.51 BLEU (2.2–2.4% EM) with gaps −1.46 to −2.63; at its final step, 11.32–11.81 BLEU with gaps −1.77 to −2.49. Neither the memorisation signature nor a positive probe response appears at any matched step, so their absence is not an artefact of the longer training horizon or the numerical-floor termination.

CTC-term normalisation.

OR1 Table S54 (OR1 Sup. V) varies the CTC denominator at fixed clip 1.0 (seed 42). None of the four candidates reproduces the released log's step-14 `Recognition Loss` (17.28; our per-sentence value 39.9, per-gloss-token 5.5, per-frame 0.74, raw sum 20,527), so the original normalisation remains unidentifiable from public artifacts. Substantively the choice is mild within the clipped regime: the gloss-token and frame denominators train slightly below the per-sentence default (dev 10.79 and 9.55) and remain signature-free, while the raw-sum variant is degenerate *at clip 1.0* (dev 0.90; the CTC-dominated gradient direction crowds the translation head out of every norm-truncated update). The degeneracy is itself a clipping artefact: with clipping disabled the same raw-sum objective trains healthily (the candidate-upstream run of §4.3, whose literal `_train_batch` semantics it reproduces).

Gradient clipping is operationally load-bearing in our reconstruction: a dose–response.

Table 2 varies only the gradient-clip threshold (seed 42; the 1.0 row is the seven-correction run of Table 1). The training-pool readout rises monotonically with the threshold—11.96 (2.4% EM) at clip 1.0, 23.65 (9.0%) at 2.0, 53.21 (31.8%) at 3.0, 82.69 (68.4%) at 5.0, and 99.98 (99.9%) unclipped—and the probe gap turns positive in lock-step on the test split (−2.80, −0.45, +5.04, +8.05, +11.19; released +10.24) and on the dev-split consistency probe (−1.35, −1.79, +4.14, +5.96, +8.07; released +8.10), where the two negative cells differ by less than typical seed dispersion and the ordering is monotone from clip 2.0 upward. The dose–response is not a seed-42 artefact: with seeds 43–44 added (three seeds per intermediate threshold), readout and the test gap are each monotone in the threshold in *every* seed, the dev-probe gap in every seed from clip 2.0 upward, and for the test gap and the dev-probe gap every adjacent threshold pair is separated by more than the between-seed range at either threshold (means ±SD and per-seed values in OR1 Table S55; e.g. test gap -1.11 ± 0.63 at clip 2.0 vs. $+5.39 \pm 0.69$ at 3.0; the seed-43/44 runs terminated at the 1,200-validation safety cap, by which point the LR was already frozen at the numerical floor and the best checkpoint unchanged). The separation criterion fails only at the top of the ladder, where the clip-5.0 and unclipped seed ranges overlap in readout (82.69–99.80 vs. 95.25–99.98): the readout saturates near the ceiling, and the two regimes are separated there by the gap and dev coordinates rather than by readout. Surface competence moves non-monotonically (dev 12.05 at 1.0, 17.17 at 2.0, 16.32 unclipped) and exceeds the released 13.38 for every threshold at or above 2.0. The released evaluator's 78.8 BLEU readout falls between clip 3.0 (53.21) and clip 5.0 (82.69)—the post-hoc closest run of the ladder, whose 68.4% EM the released 70.7% slightly exceeds—and its +10.24 gap between clip 5.0 and no clipping.

Update-level diagnostics: what clipping actually does to the optimiser.

The ladder establishes that training dynamics in our reconstruction are highly sensitive to the clipping setting; it does not by itself establish *how* the clip acts, because gradient clipping here interacts with the batch-normalised loss scale, the CTC term, Adam's first and second moments, and weight decay. Raw pre-clip gradient norms over the first ten steps (377, 568, 145, 33, ...) therefore do not imply that the Adam updates are shrunk proportionally, and instrumented runs show they are not (`src/training/train_grad_diag.py`; OR1 Sup. W). A lockstep dual run—a clip-1.0 and an unclipped replica trained from the identical seed-42 init over the identical batch order with identical dropout masks, the clipped arm replicating the published trajectory's gradient norms step-for-step (377.53, 568.21, 145.46, ...)—measures, at every optimiser step of the 2,828-step released horizon, the pre-clip gradient norm, the post-clip norm, the realised parameter-update norm $\|\theta_t - \theta_{t-1}\|$, the cosine between the update and the raw gradient, and (every 14th step) the translation and CTC gradient-component norms. The headline facts: (i) at the first step the two replicas take updates of essentially identical norm (2.42 vs. 2.42) despite the 377× gradient reduction—Adam's bias-corrected moment ratio is scale-invariant under uniform gradient rescaling, so the clip's influence is mediated by the fixed weight-decay term (no longer dominated by the gradient at post-clip norm 1.0), by the second-moment history, and by the trajectory-level rotation of update directions (mean cosine between the regimes' matched-step updates 0.45 over the first 100 steps, median 0.06 over the full window as the trajectories decorrelate); (ii) the update-norm ratio (unclipped/clipped) is 0.89 on average over the first 200 steps and 2.12 in median over the full window—the clipped run's Adam updates are *not* proportionally smaller, and over the full run they are the smaller ones (a comparison that conflates the regimes' diverging LR schedules, which we report as such); (iii) the clipped-step proportion is 100%—every step of the matched window *and* of the full 8,120-step natural-termination trajectory has a pre-clip norm above the threshold (full-trajectory median 46.4); (iv) the CTC and translation components are of comparable magnitude in both arms (medians 21.9 vs. 30.7 in the clipped arm), with CTC dominating the unclipped arm's earliest steps (38.3 vs. 12.1 at step 14). A complementary control retrains unclipped at a learning rate calibrated so that its realised update norms match the clipped run's (5.5×10^{-4}): the control *still enters the memorisation regime* (readout 84.91 BLEU / 75.1% EM, probe gap +9.51, dev 15.68 at its best checkpoint)—realised update magnitude alone does not gate the signature. We accordingly state the claim at the level the evidence supports: *training dynamics in our reconstruction are highly sensitive to the clipping setting*, with the dose–response ladder as the mechanism-level clue and the update-level diagnostics showing the effect operates through the optimiser's interaction with the clipped gradient scale—not through a simple proportional shrinkage of the parameter update.

Table 2: Gradient-clip sensitivity (seed 42; otherwise the seven-correction protocol; the 1.0 row is Table 1's seed-42 run). "none" disables clipping; "cand.-upstream" = none + CTC raw-sum normalisation (§4.3). "Dev-probe gap": the PURE−REC contrast on the 515-query dev-split consistency registry (§4.1); released evaluator +8.10. Multi-seed means in the text.

Clip threshold	Dev BLEU-4	Test REC	PURE	Test gap	Dev-probe gap	Readout (EM%)
1.0 (seven-corr.)	12.05	13.11	10.32	−2.80	−1.35	11.96 (2.4%)
2.0	17.17	16.35	15.90	−0.45	−1.79	23.65 (9.0%)
3.0	17.13	16.78	21.82	+5.04	+4.14	53.21 (31.8%)
5.0	16.54	15.92	23.97	+8.05	+5.96	82.69 (68.4%)
none	16.32	15.08	26.27	+11.19	+8.07	99.98 (99.9%)
cand.-upstream	16.04	15.12	25.80	+10.68	+7.87	97.60 (95.1%)
Released evaluator	13.38	12.78	23.02	+10.24	+8.10	78.8 (70.7%)

Source evidence: under the candidate upstream semantics, the released configuration implies no clipping.

The clip threshold was disclosed as an author inference because the released `config.yaml` specifies no clipping field. The strongest public reading of that absence comes from the upstream framework. The SLRTP2025 evaluation repository releases *no training script*; it states that the evaluation model derives from Sign Language Transformers, and its data pipeline, model definitions, and `validations.txt` report format correspond field-for-field to the signjoey code base. The specific public repository we could find whose code matches the released pipeline and whose timeline fits the challenge is `NaVi-start/Sign-IDD-SLT` (a signjoey-derived public repository under Apache-2.0; commit `249d3cd8fc249b1a06eba39f84cb2d289ed37bce`, 2025-04-27, verifiable at the repository HEAD; it is not a GitHub fork of `neccam/slt`—whose own public history ends in 2020—but an independent repo carrying the same code lineage). The commit-selection evidence (column-format and code-path correspondence) and a verbatim source archive with per-file SHA-256 hashes are in the artifact (`third_party/signjoey-signidd-slt-249d3cd8fc2/`) and in OR1 Sup. U. In that source, `build_gradient_clipper` implements both value clipping (`clip_grad_val` → `clip_grad_value_`) and norm clipping (`clip_grad_norm` → `clip_grad_norm_`; the two are mutually exclusive); when *neither* field is present the builder returns `None` and no clipping is applied. The released config contains neither field. Under the candidate upstream framework semantics, the released configuration therefore implies *no clipping*—the regime of the "none" row—and the behavioural reconstruction is consistent with this interpretation. What the public artifacts cannot establish is that the original training used exactly this framework version with no local modifications: no training transcript or organiser confirmation is available, so the attribution is conditional ("under the candidate upstream semantics"), not transcript-confirmed. The literal candidate-upstream training semantics—no clipping *and* the CTC raw sum (OR1 Sup. U)—give the candidate-upstream row, whose probe gap (+10.68) is the closest single reproduction of the released +10.24. Our 1.0 threshold—imported from the v1–v3 scripts and disclosed as an inference—was therefore an *eighth* defect in our reconstruction, and a load-bearing one: it is the single choice that hid both the readout signature and the probe response, and it did so invisibly, since every metric visible during training (dev BLEU, NLL, CTC) looks healthy at clip 1.0.

Step-matched unclipped readout, multi-seed replication, and a secondary dev-split consistency check.

The result is not a seed-42 accident or a training-duration artefact. Across seeds 42–44, unclipped runs give test-split gaps +11.19, +11.31, and +10.31 with readouts 99.98, 95.25, and 99.98 BLEU (90.5–99.9% EM) at dev 15.74–16.32. It also holds on the *dev-split consistency probe*—the 515-query donor registry constructed identically to the test probe but never used for probe construction or exploration (§4.1): the three unclipped seeds score gaps +8.07, +8.00, and +8.64 and the candidate-upstream variant +7.87, against the released evaluator's +8.10 on the same registry (`results/dev_probe_eval.json`). As noted in §4.1, this is a secondary consistency check rather than an independent confirmation, since the dev split participates in checkpoint selection and LR scheduling for every run involved. In a fresh unclipped run with checkpoints fixed at the released run's own steps, the readout is already 99.94 BLEU / 99.7% EM at step 1,820 (dev 15.69, gap +11.68) and 99.98 / 99.9% at its final step 2,828—the memorisation regime is entered *before* the released run's best checkpoint, so the overshoot relative to the released 78.8 is not an artefact of training longer than the released run. What remains unreproduced is the released operating point itself: no single knob places dev 13.38, readout 78.8, REC 12.78, and gap +10.24 simultaneously—clip 5.0 (post-hoc, selected after inspecting the ladder) matches the readout (82.69/68.4%) but overshoots dev by 3.2, while clip 1.0 matches dev-range but not the signature—so at least one further quantitative difference (CTC normalisation, stopping point, framework version, or data order) persists: the released point's *individual coordinates* are bracketed by different clipping settings, and the point itself is not matched.

Three levels of reconstruction.

The sensitivity results support three statements of increasing strength, and we claim only the first two. *(i) Qualitative signature reproduction* (supported): an implementation of the released recipe with the clip inference removed produces both previously unreproduced properties—high training-pool readout and a positive probe response—across seeds and on the never-explored dev-split consistency probe. *(ii) Partial numerical bracketing* (supported): the released evaluator's individual coordinates (dev, REC, readout, gap) each fall inside the range spanned by the clip ladder, but no single threshold matches all of them simultaneously within any prespecified tolerance, so the released operating point is bracketed coordinate-wise, not reconstructed. *(iii) Mechanism confirmation*—that the original training in fact ran unclipped—is *not* established: it rests on the candidate-upstream source semantics plus behavioural consistency (§4.3), and would require a released training transcript or organiser confirmation.

Panel-wide uniformity under the shared inference, and its reinterpretation.

Across every constructible evaluator of the canonical panel—eight seven-correction runs, fourteen partially-corrected runs, fourteen legacy seeds, and the diagnostic families, all sharing the clip=1.0 inference—the probe ordering does not reproduce: all 69 non-degenerate decoded runs have strictly negative gaps (range $[-2.80, -0.21]$; 0 positive). For the fourteen legacy reconstructions the within-implementation seed dispersion interval is $[-1.94, -0.46]$ (the six prospectively specified seeds give $[-2.48, -0.28]$, below zero in all leave-one-seed-out folds; difference-in-differences estimates +11.04 to +12.09, OR1 Table S2, every interval excluding zero); the seven-correction family shows the same sign at substantially higher competence ($[-2.80, -1.37]$). Because those runs reach the released evaluator's test-split decoding competence while lacking its training-pool signature, the non-reproduction is not attributable to surface competence; §4.3 attributes it to the one hyperparameter all 69 runs share. The 69-run uniformity is thereby reinterpreted as robustness *under a wrong inference* shared by every run, and the probe response as constructible after all (competence-gate details: OR1 Sup. D–E).

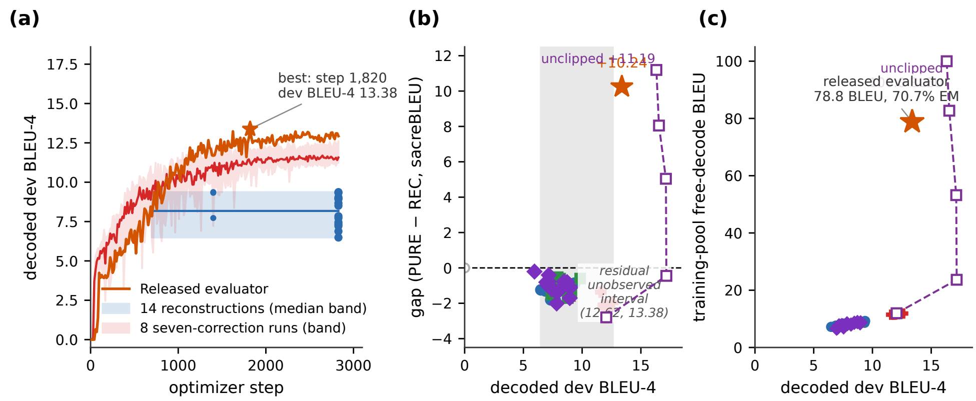

Fig. 1: Competence and training-pool readout separate the released evaluator from every canonical (clip=1.0) checkpoint. (a) Dev-BLEU-4 trajectories on a common step axis: released (orange) reaches 13.38 at step 1,820 of 2,828; defective-protocol reconstructions (blue band) plateau at 8–10, the eight seven-correction runs (red band) at 11.6–12.6. (b) Gap vs. dev BLEU-4 by family (released star); the $(12.62, 13.38)$ interval is unobserved by recipe-constructed checkpoints (the nine fine-tunes there are weight perturbations). (c) Full-pool readout vs. dev BLEU-4: canonical checkpoints read out at most 12.1 BLEU; the released evaluator 78.8 (70.7% EM); the unclipped run 99.98 (99.9% EM). Families are descriptive; checkpoints are not IID replicates.

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 1a); the defective families plateau at 8–10 (best 10.5) and the seven-correction family at 11.6–12.6, so within the canonical panel the final 0.76-BLEU ascent and the readout signature are not explained; the clipping sensitivity (§4.3) attributes both to the clip threshold. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at k=8: REC 8.75, PURE 11.69, gap +2.94; OR1 Sup. F).

The gap is positive across every decoded-text metric under the released evaluator (BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER +13.90 sign-flipped) and negative across all metrics under the six prospectively specified reconstructions, while teacher-forced NLL runs the other way (OR1 Table S32, OR1 Sup. H rank stability).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal: Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags) and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T, and Alkain et al. (2026) showed that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a *reference-frame contrast* of 9.08 BLEU points (bootstrap CI $[7.44, 10.86]$; ratio form and per-item views in OR1 Table S41, Figs. S2/S5). The interval for the residual gap includes zero (paired item CI $[-0.13, 1.99]$). This contrast is a *single-annotator, 461-item, post-hoc association*, not an intervention and not a reference-defect effect: reference replacement simultaneously changes reference content, expression style/translationese, BLEU floor level, and the lexical coupling between retrieval selection and scoring reference. Teacher-forced BT likelihood does not reproduce the probe response under either reference Jiang et al. (2025).

Scoring against both references jointly leaves the gap at +11.50, *larger* than original—adding the original reference injects a second lexically coupled target—so the response is specific to scoring frames lexically coupled to the retrieval query (full grid in OR1 Sup. T). Under all six prospectively specified reconstructions the human-reference gap is negative (−0.15 to −0.65). The human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so semantic adequacy of TN-PURE-v1 is unmeasured.

4.4 Donor-origin contrast

Restricting retrieval to unseen poses inverts the response: UNSEEN-PURE-v1 (640 other test poses) scores 8.51 vs. REC 12.78 (gap −4.27; unmatched pool-substitution contrast +14.51, OR1 Table S40). Because SEEN and UNSEEN differ in pool size, donor–reference Jaccard, donor reuse, and exposure at once, three designs bracket the confounding: Jaccard-caliper matching (148 items, SMD 0.065) leaves +9.56 $[+7.47, +11.76]$; pool-size-controlled resampling (1,000 draws of 640-donor pools) leaves +8.38 $[+7.44, +9.34]$; and a 605-query seen/unseen factorial decomposes the unmatched contrast into a pool-origin main contrast +2.08 $[+1.56, +2.57]$, a similarity contrast +6.96 $[+5.93, +8.08]$, and an interaction +3.70 $[2.63, +4.79]$ (Holm p=0.0003; across reconstructions the seen main contrast shrinks to +0.51). All three remain associations, not identified training-exposure effects (three different target populations; OR1 Sup. E).

4.5 Competitive-system context

Because the audit's adversarial probe is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: the "Ground Truth" row reproduces our canonical REC numbers exactly (BLEU-4 12.78, Walsh et al. (2025); OR1 Table S35), and the retrieval-based winner USTC-MoE sits roughly even with REC on lexical BT metrics (BLEU-4 −0.72). We re-decoded two realistic retrieval outputs under all available evaluators (OR1 Sup. P): the deployment-realistic noisy-gloss retrieval pipeline decodes at 12.65 under the released evaluator (gap −0.13 vs.

REC), and our rebuild of the winner's fixed outputs from its released intermediates decodes at 2.42 on the 497 keys those intermediates cover (gap −9.94 against the same-subset REC of 12.36; a failed reproduction, not representative of the original entry). Cross-evaluated under the full reconstructed panel (69 unique non-degenerate evaluators—the gap panel's 68 unique binaries plus the never-gap-decoded rescue run, the byte-identical pair counted once; panel accounting in OR1 Sup. P), the noisy-gloss system gives negative gaps throughout (range $[-3.22, -0.07]$), and USTC-MoE remains deeply negative ($[-9.52, -4.28]$, 62 evaluators). The +10.24 gap is specific to the constructed adversarial probe; archiving fixed per-team outputs alongside public scores is the most direct suggestion we can offer for enabling rank-stability audits.

5 Discussion

RQ1—Competence recovered first, the signature second; both were our defects. After seven implementation defects were identified and corrected (§3.4), the seven-correction family reaches dev BLEU-4 11.59–12.62 against the released 13.38 (test family mean 12.27 ± 0.46 vs. 12.78); the competence gap of the earlier "failed reconstruction" was attributable predominantly to defects in our own code—four of the seven invisible without forensic arithmetic against the released training log. The training-pool signature initially appeared to be the genuine residual; the clipping sensitivity (§4.3) closes it as well: unclipped training reproduces the signature and probe response, so the eighth defect was a wrong hyperparameter inference whose effect was invisible in every training-time metric. What remains unreproduced is the released operating point itself: different clipping settings bracket its individual coordinates, but no single threshold matches all coordinates jointly, and the canonical panel's residual dev interval $(12.62, 13.38)$ is exceeded, not filled, by the sensitivity runs. The 69-run non-reproduction was real and uniform but wrong as an attribution: one hyperparameter, shared silently by all 69 runs, was responsible. Two morals: uniformity across many runs is not evidence of artifact insufficiency when the runs share an untested inference, and the two signatures are jointly constructible properties of the released recipe, co-occurring in the unclipped regime and in the released evaluator, with the within-regime readout–gap association (ρ=−0.56) reversing across the ladder.

RQ2—Reference-frame and donor-origin sensitivity. On the matched 461-item confidence=1 subset of Czehmann et al. (2026) human back-translations, reference replacement is associated with a gap reduction to +0.95 from +10.03—a single-annotator reference-frame contrast that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects. The matched donor-pool analysis (§4.4) shows the pool-origin association persists at +9.56 under SMD<0.1 Jaccard matching; the pool-size-controlled resampling (640-vs-640) gives +8.38 (conditional simulation interval $[+7.44, +9.34]$). All three pool-origin estimates remain associations, not identified training-exposure effects.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-artifact reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal (teacher-forced likelihood does not reproduce it), and whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025). Chen et al. (2022) document a reproduction failure of a released metric with the same anatomy we find for a full evaluator: an eval-only release, underspecified training semantics, and reconstructions that fail until the semantics are pinned forensically.

Scope limitation: our reconstruction vs. public-artifact sufficiency.

Our conclusion is strictly about what our disclosed implementations construct from the public artifacts; three caveats bound the sufficiency claim. First, the released operating point is bracketed coordinate-wise, not matched, and the CTC normalisation remains unidentifiable, so residual quantitative differences of unknown sign persist. Second, the clipping resolution rests on candidate-upstream source semantics (absent fields $\Rightarrow$ no clipping), an inference about the released run's framework version rather than a released training transcript. Third, all constructions share our own data pipeline. The earlier, stronger negative claim ("the artifacts do not reproduce the signature") is withdrawn with the disclosure of the eighth defect.

Value, finding, and conclusion reproducibility.

Following the tripartite framework of Cohen et al. (2018): *values* recompute (REC 12.78, PURE 23.02, gap +10.24 regenerate from the canonical registry); *findings* reproduce across all three dimensions (surface competence; readout signature and probe response unclipped), with the released operating point bracketed coordinate-wise, not matched; *conclusions* about artifact sufficiency hold operationally, subject to the caveats of §5.

Actionable recommendations for benchmark organizers.

Three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability. *(1) An undocumented framework default can masquerade as an unreproducible property.* Because the released `config.yaml` specifies no clip fields, the framework default silently governed whether our reconstruction memorised its training pool and responded to the replay probe—and nothing visible during training distinguishes the regimes; 69 runs sharing our wrong 1.0 inference uniformly hid both properties. Benchmarks should release the exact training command and seeds, pipeline scripts beyond the manifest, a SHA-256 manifest of training tensors, intermediate checkpoints, and fixed per-team pose outputs: released numbers under-determine the code that produced them. *(2) On template-dense benchmarks, retrieval entries may benefit from donor-exclusion disclosure.* A donor-exclusion sweep is cheap and exposes the reference-coupling gap; the training-pool readout ratio (released ≈ 5.9; clip-1.0 family 0.93–1.12; unclipped ≈ 6.1) and input-side pose permutation (OR1 Sup. M) are two informative probes, with generalisation to other benchmarks untested. *(3) Reference-coupling artifacts warrant a* *supplementary reference check.* The reference-frame contrast is a single-annotator case study; template-dense benchmarks could adopt *multi-annotator* back-translations as a supplementary reference set and flag systems whose gap shifts between frames.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2026); Alkain et al. (2026); the memorisation signature is corpus-specific and does not appear in the CSL-Daily control, whose recognizer is in any case too weak for strong claims (OR1 Sup. L). Per-team pose outputs are not archived; our cross-evaluator studies decode at or below REC under all 69 canonical non-degenerate evaluators (all clip 1.0). The seven-correction family terminates at the plateau scheduler's numerical LR floor (4.4e-8) rather than the configured floor (1e-8), with best checkpoints unchanged for the final 218–338 validations; the released run itself ended mid-schedule at 101 epochs, so the two stop points are not identical—§4.3 evaluates checkpoints fixed at the released run's best and final steps and finds the conclusions unchanged. The clipping resolution rests on candidate-upstream source semantics for absent config fields and is supported behaviourally, not by a released training transcript; the CTC-term normalisation remains unidentifiable from public artifacts; and the update-level diagnostics (§4.3) quantify how the clip acts through the optimiser without settling the released run's own mechanism. The unclipped reconstruction overshoots the released operating point (readout 99.98 vs. 78.8; dev 16.32 vs. 13.38), so the exact released configuration is bracketed, not recovered; the unclipped variant is consistent across the three tested seeds (gaps +11.19/+11.31/+10.31), clip 2.0, 3.0, and 5.0 are replicated across three seeds (Table 2 shows seed 42; means ±SD and per-seed values in the text and OR1 Table S55), and the candidate-upstream variant and the update-level diagnostics are single-seed. A 2026-07-14 host compromise required post-event retraining of most checkpoint families under identical protocols, with five unrecoverable early families omitted (Data availability; OR1 Sup. D). The human back-translations are a single-annotator resource without second-annotator adjudication; the reference-frame contrast is conditional on one signer's interpretation (qualitative direction stable across confidence tiers; OR1 Table S41). The probe response is a decoded-BT-text-metric effect, not a sign-language quality judgment; without target-conditioned DGS signer evaluation, semantic adequacy of the donor motion is unmeasured—"evaluator readability" and "signing quality" are distinct constructs, and this audit measures only the former.

6 Conclusion

Correcting seven implementation defects in our own code recovered most of the released SLRTP2025 BT evaluator's surface competence (dev BLEU-4 11.59–12.62 vs. 13.38; test-decoding family mean 12.27 ± 0.46 vs. 12.78). The two properties that then still failed to reproduce across 69 constructible runs—the training-pool memorisation signature (78.8 BLEU / 70.7% exact match vs. a constructible ceiling of 12.08 / 2.5%) and the adversarial probe response (+10.24 vs. $[-2.80, -0.21]$)—proved to be blocked not by an unrecovered mechanism but by an eighth defect: our gradient-clip threshold of 1.0, imported as an author inference, whereas under the candidate upstream framework semantics the released config's absent clip fields imply no clipping (§4.3; conditional on that framework version, not transcript-confirmed). Unclipped, the same recipe reproduces both properties (readout 99.98 BLEU / 99.9% exact match; probe gap +11.19) at dev 16.32, consistent across the three tested seeds and on a never-explored dev-split consistency probe; the released operating point's individual coordinates are bracketed by different clipping settings but not matched jointly, and update-level diagnostics show the effect operates through the optimiser's interaction with the clipped gradient scale rather than proportional update shrinkage. The probe itself behaves as a case-specific adversarial unit test: its response was not observed under any text-free donor-selection rule tested, was observed only on the coupled diagonal of the retrieval-query × scoring-frame grid, and was not observed for deployment-realistic systems. The audit's lasting contributions are the forensic method (pinning released-config semantics by arithmetic against the released training log, and framework source defaults against config absences), the provenance, checkpoint-registry, and evaluator-audit infrastructure, and the cautionary mechanism it documents: an undocumented framework default can make a reproducible property look unreproducible—and can do so across sixty-nine runs at once when, as here, they silently share the same wrong inference.

Statements and Declarations

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Conflict of interest.

The authors declare no competing interests.

Data governance and ethics.

The datasets and model weights are available under research-specific terms: PHOENIX-2014T under its RWTH Aachen academic licence; the SLRTP2025 pose bundle and released BT evaluator under challenge terms; the Czehmann et al. (2026) back-translations under CC BY-NC-SA 4.0; CSL-Daily under provider-specific academic terms. The artifact is a licence composition: the root MIT Licence covers only the audit software, with per-directory LICENSE/NOTICE files documenting the remainder (including ShareAlike obligations for the Czehmann CSVs) and third-party model weights excluded from redistribution (Data availability). Pose tensors retain signer identity and biometric features; we release per-item decoded text (low re-identification risk) and a curated subset of our own trained checkpoints, but not per-signer raw poses. The original consent forms are not publicly available, so we cannot independently confirm that informant consent covers derivative replay-probe analysis; no new human-subjects data was collected. No Deaf/DGS stakeholders were involved in formulating the research questions or interpreting results: the adversarial-probe framing, reference-frame analysis, and recommendations reflect a hearing-researcher perspective on a DGS resource, and Deaf-led sign-language AI research norms should guide any follow-up Desai et al. (2024). The Czehmann et al. back-translations we build upon were produced by a Deaf fluent L2 DGS signer.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the artifact at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md (git commit `92442d8`; also deposited with DOI at https://doi.org/10.5281/zenodo.21987785 (concept DOI, resolves to the latest version; current version 10.5281/zenodo.22054861)). A claim-to-file manifest mapping each headline number to its source file and verification command is at `claim_manifest.json` (artifact root); `make check-consistency` verifies the paper's counts against the disk-scanned registry, and the L1 audit environment is captured by `requirements-audit.txt` (4 packages; the full 236-package lock is needed only for L3/L4). *Third-party distribution policy:* the artifact does not redistribute third-party model weights—for the released evaluator it ships only the bundle's metadata (config, vocabularies, training log) with SHA-256 hashes plus a download script pointing at the official challenge distribution—and PHOENIX-2014T poses are likewise not redistributed (RWTH Aachen academic licence); our own audit code is MIT-licensed, a curated subset of our own trained checkpoints is mirrored externally under the upstream data terms, and the verbatim candidate-upstream source archive is redistributed under Apache-2.0 with per-file hashes (`third_party/`). *Artifact integrity note:* a root-level host compromise on 2026-07-14 was cleaned the same day; unaffected inputs are hash-verified, most trained checkpoints were rebuilt afterwards under identical protocols (byte-identical determinism check in §4.3), five unrecoverable early families are omitted, and the host is treated as untrusted pending rebuild. The electronic supplementary material is provided as Online Resource 1 (`supplementary.pdf`); upstream dataset terms are documented in the artifact README and per-directory LICENSE/NOTICE files.

Author contribution.

Both authors contributed to the study conception, design, data analysis, and manuscript preparation, and both read and approved the final manuscript. Specific contributions (CRediT taxonomy): **Xin Ji**: conceptualisation, methodology, software, investigation, formal analysis, data curation, writing—original draft, visualisation. **Cheng Zhang**: methodology, supervision, writing—review and editing, project administration.

Electronic supplementary material (Online Resource 1)

The supplementary materials, referenced throughout the main text as OR1, are provided as a separate file (`supplementary.pdf`) and in the anonymous artifact. Each appendix is annotated with the question it answers; appendix labels match the references in the main text. **Sup. A** Missing Test ID Sensitivity; **Sup. B** Per-seed Legacy and Extension Cells; **Sup. C** Analysis Provenance and Multiplicity; **Sup. D** Canonical Checkpoint Registry and Claims Register; **Sup. E** Donor-Pool Substitution Decomposition; **Sup. F** Config-Mapping and Perturbation Details (incl. the four-column config-mapping table); **Sup. G** Cluster-Aware Bootstrap; **Sup. H** Supplementary Oracle Diagnostics; **Sup. I** Template-Density, Holdout Boundary, Donor-Exclusion; **Sup. J** Transcript-Slot Diagnostic; **Sup. K** Competence Extrapolation and Per-Item Gap Decomposition; **Sup. L** CSL-Daily Boundary Control; **Sup. M** Leakage Sanity Checks; **Sup. N** Pre-Exclusion Materialization Sensitivity; **Sup. O** BLEURT-Substitute Validation; **Sup. P** Main-Text Tables (moved from the main text); **Sup. Q** Additional Protocol Details; **Sup. R** Reconstruction Sanity Checks; **Sup. S** Dev-Split Consistency Check and Reference-Free Retrieval (incl. the full probe-multiplicity matrix); **Sup. T** Retrieval-Query Decoupling Grid; **Sup. U** Gradient-Clipping and CTC-Normalisation Sensitivity (incl. the candidate-upstream source archive and commit evidence); **Sup. V** Additional Analyses (split reconciliation, dev-split consistency panel, multi-seed clip ladder, tables relocated from the main text); **Sup. W** Update-Level Gradient Diagnostics.

Full checkpoint registry in OR1 Sup. D.

References

Alkain, B., A. Núñez-Marcos, C. Escolano, L. Docío-Fernández, O. Perez-de Viñaspre, and G. Labaka. 2026. Critical analysis of datasets for sign language translation. *Frontiers in Artificial Intelligence* 9: 1743223. https://doi.org/10.3389/frai.2026.1743223 .

Artiaga, K., S. Kamila, H. Afli, C. Lynch, and M. Hasanuzzaman 2025. Rethinking sign language translation: The impact of signer dependence on model evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China, pp. 18379–18391. Association for Computational Linguistics.

Camgoz, N.C., S. Hadfield, O. Koller, H. Ney, and R. Bowden 2018. Neural sign language translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7784–7793.

Carlini, N., F. Tramer, E. Wallace, M. Jagielski, A. Herbert-Voss, K. Lee, A. Roberts, T. Brown, D. Song, U. Erlingsson, A. Oprea, and C. Raffel 2021. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650.

Chen, Y., J. Belouadi, and S. Eger 2022. Reproducibility issues for BERT-based evaluation metrics. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2965–2989. Association for Computational Linguistics.

Cohen, K.B., J. Xia, P. Zweigenbaum, T.J. Callahan, O. Hargraves, F. Goss, N. Ide, A. Névéol, C. Grouin, and L.E. Hunter 2018. Three dimensions of reproducibility in natural language processing. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, Miyazaki, Japan, pp. 156–165. European Language Resources Association (ELRA).

Cory, O., M. Ivashechkin, K. Sahin, O. Ranum, J. Low, E. Fish, A. Pelykh, O. Mercanoglu Sincan, and R. Bowden. 2026. BackTranslation2.0 – a linguistically motivated metric to assess sign language production. *arXiv preprint arXiv:2606.28673*. arXiv:2606.28673 .

Czehmann, V., S. Yazdani, Y. Hamidullah, F. Nunnari, and E. Avramidis 2026. "A Sacred Bird Called the Phoenix": Auditing the most-used parallel corpus for German sign language recognition and translation. In *Proceedings of the LREC2026 12th Workshop on the Representation and Processing of Sign Languages: Language in Motion*, Palma, Mallorca, Spain, pp. 80–92. ELRA Language Resources Association. ISBN 978-2-493814-82-1. Back-translated test set and annotations: https://github.com/DFKI-SignLanguage/sacre-bird-phoenix (CC BY-NC-SA 4.0).

Desai, A., M. De Meulder, J.A. Hochgesang, A. Kocab, and A.X. Lu. 2024. Systemic biases in sign language AI research: A deaf-led call to reevaluate research agendas. *arXiv preprint arXiv:2403.02563*. arXiv:2403.02563 [cs.CV].

He, T., K. Rahul, D. Hupkes, and R. Cotterell 2024. On the blind spots of model-based evaluation metrics for text generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics.*

Hong, R. and Košecká, Jana. 2026. Conditional collapse in sign language production: A diagnostic and a scaling argument. *arXiv preprint arXiv:2606.01643*. arXiv:2606.01643 [cs.CV].

Imai, S., M. Inan, A.B. Sicilia, and M. Alikhani 2025. SiLVERScore: Semantically-aware embeddings for sign language generation evaluation. In *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, pp. 452–461.

Jiang, Z., C. Leong, A. Moryossef, O. Cory, M. Ivashechkin, N. Tarigopula, B. Zhang, A. Göhring, A. Rios, R. Sennrich, and S. Ebling 2025. Meaningful pose-based sign language evaluation. In *Proceedings of the Tenth Conference on Machine Translation (WMT)*, pp. 64–80.

Kandpal, N., E. Wallace, and C. Raffel 2022. Looking inside my memory: Quantifying memorization in language models. In *Findings of EMNLP.*

Khandelwal, U., A. Fan, D. Jurafsky, L. Zettlemoyer, and M. Lewis 2020. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR).*

Kim, J.H., M. Huerta-Enochian, C. Ko, and D.H. Lee 2024, May. SignBLEU: Automatic evaluation of multi-channel sign language translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italia, pp. 14796–14811. ELRA and ICCL.

Post, M. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pp. 186–191.

Saunders, B., N.C. Camgoz, and R. Bowden 2020. Progressive transformers for end-to-end sign language production. In *European Conference on Computer Vision*.

Saunders, B., N.C. Camgöz, and R. Bowden. 2021. Continuous 3D multi-channel sign language production via progressive transformers and mixture density networks. *International Journal of Computer Vision* 129: 2113–2135. https://doi.org/10.1007/s11263-021-01457-9 .

Tasyurek, S.M., T. Kiziltepe, and H.Y. Keles. 2025. Disentangle and regularize: Sign language production with articulator-based disentanglement and channel-aware regularization. *arXiv preprint arXiv:2504.06610*. arXiv:2504.06610 .

Walsh, H., E. Fish, O. Mercanoglu Sincan, M.I. Lakhal, R. Bowden, N. Fox, B. Woll, K. Wu, Z. Li, W. Zhao, H. Wang, W. Zhou, H. Li, S. Tang, J. He, X. Wang, R. Zhang, Y. Wang, L. Cheng, M. Tasyurek, T. Kiziltepe, and H.Y. Keles 2025. SLRTP2025 sign language production challenge: Methodology, results, and future work. In *Proceedings of the Third Sign Language Recognition, Translation and Production Workshop at CVPR 2025*. arXiv:2508.06951.

Walsh, H., A. Ravanshad, M. Rahmani, and R. Bowden 2024. A data-driven representation for sign language production. In *British Machine Vision Conference*.

Walsh, H., B. Saunders, and R. Bowden 2024. Sign stitching: A novel approach to sign language production. In *British Machine Vision Conference*.

Yazdani, S., Y. Hamidullah, C. España-Bonet, E. Avramidis, and J. van Genabith 2026. A critical study of automatic evaluation in sign language translation. In *Proceedings of the 15th International Conference on Language Resources and Evaluation (LREC 2026)*, pp. 9535–9548.

Zhang, T., V. Kishore, F. Wu, K.Q. Weinberger, and Y. Artzi 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*.

Zuo, R., R.A. Potamias, E. Ververas, J. Deng, and S. Zafeiriou 2025. Signs as tokens: A retrieval-enhanced multilingual sign language generator. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.